

Engineering Assumptions

Since the published research paper did not provide all the operating specifications required to completely reproduce the extractive distillation process, several engineering assumptions were made during process development. These assumptions were based on chemical engineering principles, convergence behaviour, published literature, and systematic sensitivity analysis rather than arbitrary selection.

Assumption 1 – Missing Operating Parameters

The reference paper did not report all process specifications required for complete simulation (such as certain column operating conditions, convergence specifications, and some operating variables).

Therefore, the missing parameters were estimated using engineering judgement and subsequently refined through systematic convergence studies until a physically realistic and stable base case was obtained.

Assumption 2 – Thermodynamic Model Selection

The following thermodynamic models were adopted for the simulation:

- Activity Coefficient Model : NRTL
- Vapour Phase Model : Peng–Robinson Equation of State
- Property Method : Gamma–Phi
- Vapour Pressure Correlation : Extended Antoine

--Reason

The ethanol–water system exhibits strong non-ideal liquid behaviour due to hydrogen bonding and azeotrope formation. The NRTL model is widely accepted for modelling highly non-ideal mixtures and extractive distillation systems involving polar components.

Peng–Robinson was selected for vapour-phase calculations because it provides stable convergence and reliable vapour property estimation within ChemSep.

Assumption 3 – Physically Realistic Operation

Only operating conditions producing physically meaningful results were considered.

Simulation cases resulting in unrealistic temperatures, unstable material balances, or failure to converge were rejected from further analysis.

Assumption 4 – Converged Cases Only

Only fully converged simulation cases were included for dataset generation.

Non-converged simulations were excluded from machine learning model development to ensure that the AI learned only from physically feasible operating conditions.

Assumption 5 – Sensitivity-Based Operating Ranges

The operating ranges used throughout this work were determined through systematic sensitivity analysis.

Instead of selecting arbitrary parameter limits, each variable was varied until the practical convergence limits of the process were identified. The final operating ranges were selected from these stable convergence regions.

Assumption 6 – Independent Evaluation of Process Variables

Although several operating conditions were reported in the reference paper (for example, the reflux ratio of Column 2), these parameters were intentionally varied during this study.

--Reason

The objective of this work was not only to reproduce the published results but also to investigate the influence of each operating parameter on:

- Ethanol purity
- Ethanol recovery
- Ethylene glycol recovery
- Energy consumption
- Overall Process Performance Index (PPI)

This enabled the generation of a comprehensive engineering dataset suitable for machine learning model development and process optimization.

Assumption 7 – Validation Base Case

A validated operating case was established after completing the convergence studies.

This validated base case was used as the reference operating condition for all subsequent sensitivity analyses, dataset generation, machine learning model development, and AI-based optimization.

Assumption 8 – Machine Learning Dataset

The machine learning models were trained exclusively using converged simulation data.

Consequently, the AI optimization framework predicts operating conditions only within the physically meaningful process domain identified through engineering analysis.

Assumption 9 – Fixed Process Configuration

The overall process configuration was maintained constant throughout the study to ensure a fair comparison between different operating conditions.

The following process parameters were kept fixed:

- Process flowsheet configuration
- Number of distillation columns
- Solvent type (Ethylene Glycol)
- Feed composition
- Feed stream arrangement
- Number of stages in Column 1
- Feed stage and solvent stage locations in Column 1
- Number of stages in Column 2
- Feed stage location in Column 2
- Column operating pressures (except during dedicated sensitivity studies)

Only the selected operating variables identified through sensitivity analysis—such as boilup ratio, bottom flow rate, reflux ratio, solvent temperature, and pressure during pressure studies—were varied.

--Reason

Keeping the process configuration constant ensured that any observed improvements in separation performance or Process Performance Index (PPI) were solely due to optimization of operating conditions rather than modifications to the process design itself. This approach provides a fair engineering comparison and mirrors the optimization strategy commonly adopted in industrial process improvement studies.